Israel Gebresilasie Kimo

Passport: EP8527470 **Nationality:** Ethiopian **Date of birth:** 26/07/1995

Place of birth: Arba Minch **Phone number:** (+251) 927833912

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Work: Arba Minch Ethiopia, 21 Arba Minch (Ethiopia)

ABOUT ME

I am a lecturer and researcher in Meteorology and Earth System Science with expertise in atmospheric dynamics, numerical weather prediction, atmospheric thermodynamics, climate change, atmospheric modeling, and environmental data analysis. I have programming skills in Python and R, with experience in scientific computing, statistical analysis, visualization, and the management and analysis of large meteorological and environmental datasets.

My research experience includes atmospheric modeling, atmospheric transport, climate change, and air quality studies, resulting in peer-reviewed publications in international journals indexed in Scopus and Web of Science. Recently, I have expanded my research focus to machine-learning applications for weather prediction and am currently working on on machine-learning-based rainfall prediction in Ethiopia.

I have extensive teaching experience in Atmospheric Dynamics, Numerical Weather Prediction, Earth System Science, Atmospheric Thermodynamics, Climate Change, and related meteorological courses. Examples of my teaching materials are available at: https://israelgebresilasiekimo.github.io/aam-micrometeorology/teaching.html. I also maintain GitHub repositories showcasing my programming and data-analysis projects: https://github.com/IsraelGebresilasieKimo.

I work effectively both independently and within multidisciplinary research teams, demonstrating strong analytical, problem-solving, and communication skills. I have experience presenting research findings at academic conferences and collaborating on interdisciplinary research projects. In addition, I serve as a reviewer for international scientific journals and have published peer-reviewed articles in Scopus- and Web of Science-indexed journals.

I am highly motivated to further develop my expertise in machine learning, explainable artificial intelligence, high-performance computing, and data-driven weather forecasting. With my interdisciplinary background, strong quantitative skills, and commitment to scientific research, I am well prepared to contribute to innovative research at the interface of atmospheric science and artificial intelligence.

WORK EXPERIENCE

Arba Minch University – Arba Minch, Ethiopia

Lecturer and Researcher

[28/07/2018 – Current]

EDUCATION AND TRAINING

MSc in Climate Change and Development

Arba Minch University [30/08/2018 – 28/07/2021]

City: Arba Minch | **Country:** Ethiopia | **Website:** www.amu.edu.et

BSc In Meteorology and Hydrology

Arba Minch University [30/08/2015 – 25/06/2018]

City: Arba Minch | **Country:** Ethiopia | **Website:** www.amu.edu.et

Basic Python for Environmental Data Processing Training of Trainers (ToT)
Arba Minch University [18/09/2022 – 30/10/2022]

City: Arba Minch | **Country:** Ethiopia | **Website:** www.amu.edu.et

LANGUAGE SKILLS

Mother tongue(s): Amharic

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Machine Learning • / Phyton / computational fluid dynamics / R / Geographical Informational Systems (GIS) / Organizational and planning skills / great knowledge of tools like excel, word, power point

PUBLICATIONS

[2026]
SPS30 and SEN55 PM2.5 sensor intercomparison and validation through indoor and outdoor measurements in Arba Minch, Ethiopia Preprint

Journal Name: EGUsphere | **Publisher**: EGUsphere (Copernicus Publications / European Geosciences Union preprint server)

[2025]
Exploring climate change adaptation pathways for the agricultural sector in Arba Minch Zuria and Bonke districts: Based on CCAFS climate analogue tool

Journal Name: Climate Services | **Publisher**: Elsevier

[2024]
Identifying the Moisture Sources in Different Seasons for Abaya-Chamo Basin of Southern Ethiopia Using Lagrangian Particle Dispersion Model

PROJECTS

[31/12/2025 – 06/05/2026]
My Scientific Web Project on Micrometeorology and Weather-Resilient Advanced Air Mobility (AAM) showcases my ongoing work, simulations, and reflections on AAM safety, urban aerodynamics, and atmospheric flow modeling.

Link: https://israelgebresilasiekimo.github.io/aam-micrometeorology/